

Integration of Vienna Rectifier and SC-PFM for Soft-Switching Current Sensorless PFC With Simplified Control

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Abstract—This article presents a study on the interleaved Vienna Rectifier with pulse-frequency modulation (IVR-PFM), an integration of the Vienna Rectifier and the switching cell with pulse-frequency modulation (SC-PFM), for three-phase three-wire applications. The IVR-PFM retains key advantages of the SC-PFM, including soft switching, voltage-follower behavior, and simplified control without current sensing. Unlike converters in discontinuous conduction mode (DCM), the IVR-PFM achieves continuous input current with peak currents in active components only slightly exceeding the input current—behavior associated with continuous conduction mode (CCM). Control requires only output voltage feedback and a simple outer-loop PI controller, eliminating the need for processor-intensive control algorithms such as $dq0$ or $\alpha\beta$ transformations, space vector pulse width modulation (SVPWM), phase-locked loops (PLLs), and cosine generation. The article also analyzes the commutation behavior, input current ripple, zero-crossing dynamics, and natural voltage balancing between the dual output capacitors. Experimental results from a 3 kW prototype, operating from a 220 V line input to an 800 V output, confirm the converter's performance under varying load conditions and validate the proposed analytical models.

Index Terms—AC–DC power converters, current sensorless, power factor correction (PFC), switching converters, three-phase electric power, Vienna Rectifier, voltage control.

I. INTRODUCTION

IN recent years, the demand for high-power converters has been on the rise, driven by the increasing need for efficient power conversion in various applications, such as telecommunications systems, electric vehicle battery chargers, dc power supplies, UPS, interface of smart grid, and renewable energy systems [1], [2], [3], [4], [5], [6].

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As a result, researchers and engineers have been exploring new topologies and techniques to achieve better performance, higher power density, greater efficiency, and lower cost. One of the main areas of interest is power factor correction (PFC), essential for achieving high-quality power conversion and complying with international power quality standards.

For three-phase, three-wire converters, achieving quality standards typically involves sophisticated modulation and control techniques, including $\alpha\beta$ transformations, $dq0$ transformations, and space vector pulse width modulation (SVPWM) strategies [6], [7], [8], [9], [10], [11], [12]. Unlike the simpler approach possible with four-wire configurations where each phase can be treated independently three-wire systems require a unified approach to maintain balanced phase operation. Though effective, these methods often demand considerable computational resources, especially in applications that prioritize real-time performance and adherence to power quality standards.

Soft-switching methods such as zero-voltage switching (ZVS), zero-current switching (ZCS), and zero-voltage ZCS (ZVZCS) effectively reduce switching losses and thermal stress by ensuring minimal voltage or current at the switching instant. While some implementations for three-phase systems require additional components, others achieve these transitions through optimized control strategies alone, enhancing efficiency in high-power applications [1], [13], [14].

The Vienna Rectifier, another topology with common use in high-power converters, is a unidirectional three-level boost rectifier known for its low total harmonic distortion (THD), unit power factor, high power density, and high efficiency [12], [15], [16], [17], [18]. The conventional configuration, originally proposed in [19], is illustrated in Fig. 1(a).

A promising approach for PFC is the switching cell with pulse-frequency modulation (SC-PFM) topology, which offers, among many advantages, soft commutations, continuous input current, and a resistive characteristic. Moreover, SC-PFM converters operate in interleaved mode, improving overall performance and efficiency. Fig. 1(b) presents the SC-PFM adapted for integration with the Vienna Rectifier.

In this article, we explore the integration of the SC-PFM with the Vienna Rectifier in a three-phase, three-wire configuration, resulting in the interleaved Vienna Rectifier with pulse-frequency modulation (IVR-PFM). This arrangement combines

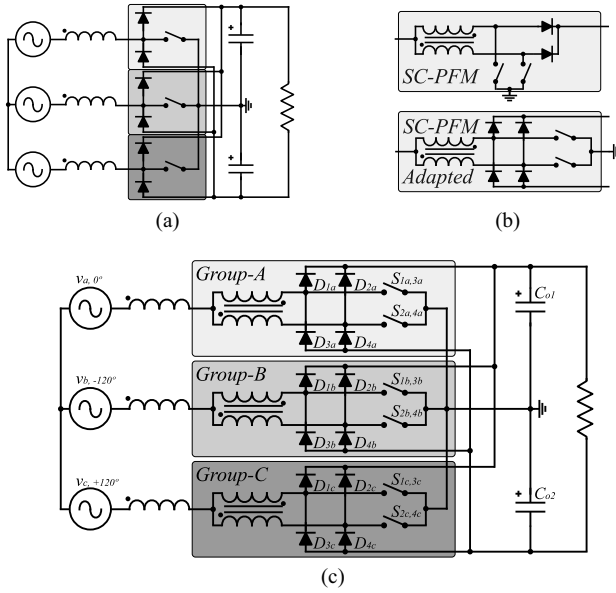


Fig. 1. (a) Conventional Vienna Rectifier [19]; (b) SC-PFM adapted for integration with the Vienna Rectifier [20], [21]; and (c) IVR-PFM topology: each input connection will have a set of components. Each bidirectional switch represents two MOSFETs connected in opposition to each other. The converter operates with a fixed duty cycle of 50% using pulse-frequency modulation (PFM). The higher the switching period, the greater the power conversion.

the benefits of both topologies and is particularly suited for high-power applications. The combined advantages of this configuration include.

- 1) Input current follows the input voltage with a sinusoidal waveform and theoretical unitary power factor, with only minor disturbances near the zero-crossing instants.
- 2) Soft-switching behavior over a wide power range, including ZVS and ZVZCS transitions, with diode blocking occurring at zero current—eliminating reverse recovery losses.
- 3) High voltage gain achieved by the SC-PFM stage, offering twice the gain of a conventional boost converter, combined with the Vienna Rectifier's dual-capacitor structure that further elevates the output voltage.
- 4) No need for current sensors (no current loop is required).
- 5) AC input voltage sensing can be simplified to just half-line cycle detection.
- 6) No need for synchronous reference frame transformations ($dq0$, $\alpha\beta$), SVPWM, phase-locked loop (PLL), frequency-locked loop (FLL), second-order generalized integrator (SOGI), and cosine signal calculations; the control strategy is simple and avoids processor-intensive computations.
- 7) The circuit topology inherently ensures balanced voltages across the two output capacitors, eliminating the need for dedicated balancing control.

Specifically, we focus on the three-phase three-wire connection, explaining its peculiarities and providing a detailed analysis of its operating principle and switching transition behavior.

We verify the concept through practical demonstration in a 3 kW, 800 V output application.

II. CIRCUIT OPERATION

The converter analyzed in this work builds upon the switching concept introduced with the SC-PFM topology, originally proposed in [20] in the context of dc–dc converters. That work introduced a soft-switching cell based on pulse-frequency modulation, featuring voltage-follower behavior, and simplified control without current sensing.

Subsequently, in [21], the SC-PFM was integrated with a Vienna Rectifier structure for a single-phase ac–dc application. The same study also presented, through simulation, a three-phase, four-wire extension using three independent single-phase IVR-PFM converters connected line-to-neutral, benefiting from phase decoupling and structural simplicity.

The present work, in contrast, investigates the adaptation of the SC-PFM structure to a three-phase, three-wire configuration based on the Vienna Rectifier. Unlike four-wire implementations, the three-wire structure lacks a neutral connection and relies on a shared reference point, introducing interdependence between the phase groups. This affects current circulation, conduction intervals, and the structure of the switching states. Although subtle from a structural perspective, these differences result in distinct waveform behavior and require a dedicated structural and operational analysis, which is developed throughout this article.

To implement this configuration, the SC-PFM switching cell is integrated into each phase of the Vienna Rectifier topology. Fig. 1(a) shows the conventional Vienna Rectifier with its standard unidirectional switching cell. Fig. 1(b) presents two versions of the SC-PFM: the original cell proposed in [20], and an adapted variant compatible with the Vienna bridge structure. The resulting converter, illustrated in Fig. 1(c), integrates the SC-PFM's voltage-follower behavior and commutation benefits with the three-phase interface and structural arrangement of the Vienna Rectifier.

While the SC-PFM's core characteristics—such as soft switching, voltage-follower behavior, and simplified control—are retained, their application in a three-phase, three-wire configuration introduces new structural dynamics. In particular, the interdependence between phase groups during conduction intervals leads to modified commutation timing and current circulation behavior, which differ from both the single-phase and previously simulated multiphase cases.

In addition, this work presents several developments not addressed in prior studies. These include a detailed analysis of the commutation intervals adapted to the Vienna Rectifier structure, the characterization of the input current ripple profile, the derivation of an approximate small-signal model, and the formulation of an exponential decay model describing the natural voltage self-balancing behavior between the output capacitors.

To assist with clarity throughout the analysis, Table I lists the main components and variables used in the circuit, along with their corresponding descriptions.

The circuitry of the IVR-PFM, illustrated in Fig. 1(c), consists of three identical groups. Each group contains four

TABLE I
COMPONENTS AND VARIABLES DESCRIPTION

Component/Variable	Meaning	Component/Variable	Meaning
$^a S_{1x,2x,3x,4x}$	Switches, each converter group has four of them	$^a i_{s1x,s2x}, \dots$ $^a v_{s1x,s2x}, \dots$	Switches current and reverse voltage
$^a D_{1x,2x,3x,4x}$	Diodes, each converter group has four of them	$^a i_{d1x,d2x}, \dots$ $^a v_{d1x,d2x}, \dots$	Diodes current and reverse voltage
$^a C_{s1x,s2x}, \dots = C_s$ $^a C_{d1x,d2x}, \dots = C_d$	Parasitic capacitance of switches and diodes, each component has its own capacitance effect	$^a i_{C s1x,C d1x}, \dots$	Current that charges and discharges the components intrinsic capacitance
$^a L_x$	Input inductors and inductance value	$^a i_{Lx}$ $^a v_{Lx}$	Input inductors current and voltage
$^a ATR_x$	Autotransformers	v_{ABx}	Voltage over point A and B
$^b 4L_{mx}$	ATR Equivalent representation of the magnetizing inductance	$^a i_{4L_{mx}}$	Theoretical current flowing through $4L_{mx}$
v_{an}, v_{bn}, v_{cn}	Input phase voltage	V_p	Input phase voltage peak
C_{o1}, C_{o2}	Output capacitors and capacitance values	$v_o = v_{o1} + v_{o2}$	Output voltage equals the output capacitors voltage sum
$v_e(t)$	Unbalanced voltage over capacitors $v_{o1} - v_{o2}$	$^a G_{1x,2x,3x,4x}$	Gate command signals
$D=0.5$	Duty cycle is fixed at 0.5	f_s, T_s	Switching frequency and period
t_x	Its the time where the diodes conducts, and also the time where the $i_{4L_{mx}}$ variate	t_{c1}, t_{c2}	Transition times when the diodes blocks and the switches blocks respectively
I_{fw}	Free-wheeling current created by the resonant tank $4L_m$ and $C_s + C_d$	θ	Zero-crossing dead zone, phase angle where current remains zero at the zero-crossing boundaries
ΔI_L	Input current ripple	$\Delta I_L = \frac{\Delta I_H L}{V_p T_s}$	Parameterized input current ripple

Note: ^ax represents which phase/group it belongs, if A, B or C; ^bsee appendix.

switches ($S_{1x,2x,3x,4x}$), four diodes ($D_{1x,2x,3x,4x}$), one input inductor (L_x), and an autotransformer (ATR_x) with magnetizing inductance L_{mx} . This design employs a common midpoint, creating dual output voltage levels at v_{o1} and v_{o2} , and requires two output filter capacitors, C_{o1} and C_{o2} .

The converter functions as a full-wave rectifier, with each component playing a specific role during each half-line cycle. In the positive half-line cycle, diodes $D_{1x,2x}$ transfer energy to capacitor C_{o1} , while diodes $D_{3x,4x}$ remain reverse-biased. Switches $S_{1x,2x}$ are modulated to drive the converter's operation, while switches $S_{3x,4x}$ remain activated until the half-line cycle changes, at which point their roles are reversed. The modulation of switches $S_{1x,2x}$ is performed with a fixed duty cycle of 50%, operating in a complementary manner. The control variable for this modulation is the switching period.

For clarity, the magnetizing inductance is modeled as an inductor with inductance $4L_m$ in parallel with both autotransformer connections (points A and B), rather than a single inductor in parallel with one winding. Details and justification are in Appendix A.

The IVR-PFM exhibits behavior analogous to Ohm's law, where the input voltage (v_i) and current (i_i) are linearly dependent through the converter's equivalent resistance (R_e). This relationship, defined as a function of the switching frequency (f_s) and magnetizing inductance (L_m), has been extensively studied in prior works [20], [21], [22], [23], [24] and is expressed in (1). Both L_m and f_s significantly influence the interaction between v_i and i_i in the SC-PFM converter

$$v_i = R_e i_i = (4L_m f_s) i_i, \quad R_e = 4L_m f_s. \quad (1)$$

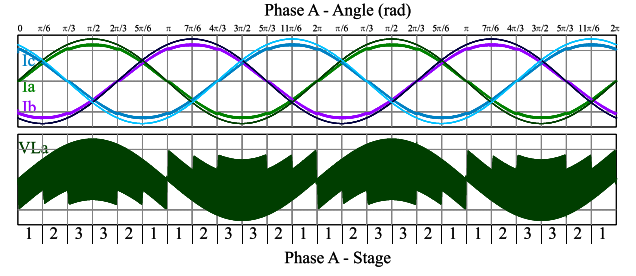


Fig. 2. Input inductor voltage stages. The division between stages is highlighted, showing the distinct operational phases within the line cycle for phase-A. Phases B and C follow a similar pattern but are offset according to their respective phase angles.

A. Principle of Operation

The operational dynamics of the three-phase, three-wire IVR-PFM significantly differ from those of the previously discussed four-wire systems, primarily due to the interactive effects between the phases which cannot be simplified to single-phase behavior. In this section, we explore the operation of one of the groups in a single phase (Phase-A) and delineate how it functions through different stages within a half-line cycle, contingent on the relative voltages of the other phases.

During the positive half-line cycle of Phase-A ($v_{an} = V_p \sin(\omega t)$), the operation of Group-A is influenced by the instantaneous voltages of Phase-B ($v_{bn} = V_p \sin(\omega t - 2\pi/3)$) and Phase-C ($v_{cn} = V_p \sin(\omega t + 2\pi/3)$). This interaction defines three operational stages, determined by the relative voltage magnitudes of each phase, as shown in Fig. 2 and Table II. These stages affect the conduction times of the components in each group.

- 1) Stage 1: $\omega t \in (0, \pi/6), (5\pi/6, 7\pi/6), (11\pi/6, 2\pi)$
where $|v_{an}| < |v_{bn}|$ and $|v_{an}| < |v_{cn}|$.
- 2) Stage 2: $\omega t \in (\pi/6, \pi/3), (2\pi/3, 5\pi/6), (7\pi/6, 4\pi/3), (5\pi/3, 11\pi/6)$
where $|v_{an}|$ is between $|v_{bn}|$ and $|v_{cn}|$.
- 3) Stage 3: $\omega t \in (\pi/3, \pi/2), (\pi/2, 2\pi/3), (4\pi/3, 5\pi/3)$
where $|v_{an}| > |v_{bn}|$ and $|v_{an}| > |v_{cn}|$.

Each group operates according to these stages, with Phase-A following the sequence 1-2-3, mirrored in Phases B and C with a phase shift.

- 1) Phase-A: 1-2-3-3-2-1.
- 2) Phase-B: 3-3-2-1-1-2.
- 3) Phase-C: 2-1-1-2-3-3.

For example, when Phase-A is in stage 1, Phases B and C are in stages 3 and 2, respectively. This cyclic pattern repeats throughout the line cycle.

B. Voltage Follower Characteristics

This converter exhibits resistive behavior with voltage follower characteristics, where the input current is directly proportional to the input voltage. This proportionality is governed by the magnetizing inductance of each autotransformer group and the switching frequency.

TABLE II
WAVEFORMS AND TOPOLOGICAL STATES

Waveforms	Topological States			
	Topological State 1-1: $t_0 < t < t_a$	Topological State 1-2: $t_a < t < t_c$	Topological State 1-3: $t_c < t < t_b$	Topological State 1-4: $t_b < t < T_s/2$
	Topological State 2-1: $t_0 < t < t_c$	Topological State 2-2: $t_c < t < t_a$	Topological State 2-3: $t_a < t < t_b$	Topological State 2-4: $t_b < t < T_s/2$
	Topological State 3-1: $t_0 < t < t_c$	Topological State 3-2: $t_c < t < t_b$	Topological State 3-3: $t_b < t < t_a$	Topological State 3-4: $t_a < t < T_s/2$

The following assumptions are made for the analysis.

- 1) The input current ripple during a switching cycle is small, allowing the current through the input inductor to be approximated as constant.
- 2) The output voltage ripple is negligible, maintaining a constant output voltage v_o . The output voltage levels v_{o1} and v_{o2} are balanced, such that $v_{o1} = v_{o2} = v_o/2$.
- 3) The input currents are balanced, satisfying the condition $i_{La} + i_{Lb} + i_{Lc} = 0$.
- 4) The magnetizing inductances values are approximately equal and denoted as L_m ($L_{ma} = L_{mb} = L_{mc} = L_m$).
- 5) During a switching cycle, the input inductor voltage pattern in the second half mirrors the first half, with the operation differing only in the conducting switches and diodes.

To derive the input current equation, the voltage across the input inductor is analyzed by examining the conduction time of each diode, which corresponds to the magnetizing current duration in the autotransformers.

When a diode conducts, it carries the current through its respective autotransformer winding. The magnetizing inductance absorbs this current until it matches the winding current, at which point the diode ceases conduction. Table II details the conduction times t_a , t_b , and t_c for Groups A–C, respectively.

These times can be derived from the magnetizing current equation for Group-A

$$i_{4Lma}(t_a) = i_{4Lma}(t_0) + \frac{1}{4L_m} \int_{t_0}^{t_a} \frac{-v_o}{2} dt. \quad (2)$$

Evaluating the integral gives, considering $i_{4Lma}(t_0) = i_{La}/2$ and $i_{4Lma}(t_a) = -i_{La}/2$

$$t_a = \frac{8L_m|i_{La}|}{v_o}. \quad (3)$$

Similarly, the equations for t_b and t_c are as follows:

$$t_b = \frac{8L_m|i_{Lb}|}{v_o}, t_c = \frac{8L_m|i_{Lc}|}{v_o}. \quad (4)$$

Assuming a small input current ripple, the mean value of the input inductor voltage is zero. For Group-A in stage 1, the input inductor voltage is given as follows:

$$v_{La} = \begin{cases} v_{an} - \frac{v_o}{6}, & t_0 \leq t < t_a \\ v_{an}, & t_a \leq t < t_c \\ v_{an} - \frac{v_o}{12}, & t_c \leq t < t_b \\ v_{an}, & t_b \leq t < \frac{T_s}{2}. \end{cases} \quad (5)$$

Consequently, the diode conduction times account for I_{fw}

$$\begin{aligned} t_a &= \frac{8L_m(|i_{La}| + I_{fw})}{v_o} \\ t_b &= \frac{8L_m(|i_{Lb}| + I_{fw})}{v_o} \\ t_c &= \frac{8L_m(|i_{Lc}| + I_{fw})}{v_o}. \end{aligned} \quad (12)$$

3) Switching Block Transition: When the switches block, the intrinsic capacitor of the switch is charged, and the intrinsic capacitor of the diodes is discharged, enabling ZVS at the moment the diodes begin conducting, as shown in Fig. 3(c). The I_{fw} adds to the input current during this transition, ensuring it remains fast and effective even at low input current levels. This rapid transition, does not significantly alter the current equation for simplicity, see transition time t_{c2}

$$t_{c2} = v_o \frac{C_s + C_d}{i_{Lx} + I_{fw}}. \quad (13)$$

D. Input Current Modeling Under Nonidealities

The input current is obtained by evaluating the average voltage across each input inductor over one switching cycle, taking into account the effects previously discussed. As the converter operates through three distinct stages within each quarter of the line cycle, the current expression must be derived separately for each stage. In addition, due to nonidealities, the current waveform includes distortion intervals: a nonconducting region around the voltage zero-crossing and a flat region during the transition between stages 2 and 3.

1) Stage-1 Current and Zero-Crossing Dead Zone ($\theta < \omega t < \pi/6$): By solving the mean voltage condition for stage-1 (10) with the conduction times (12) and the free-wheeling current definition (11), the current for Group-A is given by the following equation:

$$i_{La} = \frac{v_{an}}{4L_m f_s} - \frac{4I_{fw}}{3} = \frac{v_{an}T_s}{4L_m} - \frac{4I_{fw}}{3}. \quad (14)$$

However, this expression is valid only when the first term exceeds the second, i.e., when

$$\frac{v_{an}}{4L_m f_s} > \frac{4I_{fw}}{3}. \quad (15)$$

In the vicinity of the input voltage zero-crossing, this condition is not satisfied, and the current remains zero. This nonconducting interval is referred to as the zero-crossing dead zone, whose angular extent is denoted by θ , defined as follows:

$$\begin{aligned} \theta &= \arcsin\left(\frac{16I_{fw}L_m f_s}{3V_p}\right) \approx \frac{16I_{fw}L_m f_s}{3V_p} \\ \theta &\approx \frac{4}{3} \cdot \frac{v_o}{V_p} \cdot f_s \sqrt{L_m(C_s + C_d)}. \end{aligned} \quad (16)$$

2) Stage-2 Current ($\pi/6 < \omega t < \pi/3 - \theta$): For stage 2, which occurs after the angle $\omega t = \pi/6$, the current equation remains similar to stage 1, as shown in (14). Although the relative voltage relationships of the phases differ in this stage, the evaluation of the mean input inductor voltage still leads to the same result for the stage 2 input current equation.

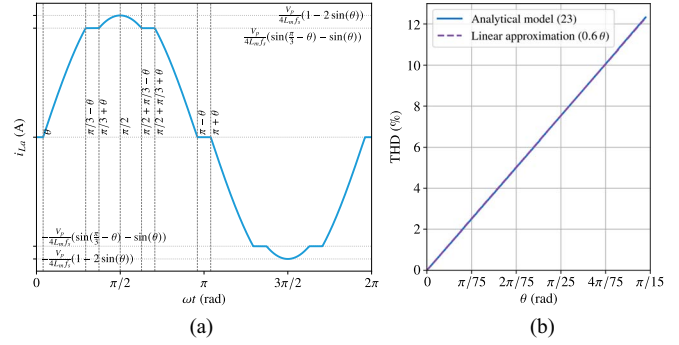


Fig. 4. (a) Input current waveform (20); and (b) estimated THD as a function of θ . The analytical result from (23) and approximation THD $\approx 0.6\theta$.

3) Transition Interval Between Stages-2 and -3 ($\pi/3 - \theta < \omega t < \pi/3 + \theta$): During this transition period, the input current of phase-C reaches zero, and phase-A's current stabilizes at a constant value

$$i_{La} = \frac{V_p \sin(\pi/3 - \theta)}{4L_m f_s} - \frac{4I_{fw}}{3}. \quad (17)$$

4) Stage-3 Current ($\pi/3 + \theta < \omega t < \pi/2$): After the transition region, stage-3 begins, and phase-A's current is given by the following equation:

$$i_{La} = \frac{v_{an}}{4L_m f_s} - \frac{8I_{fw}}{3} = \frac{v_{an}T_s}{4L_m} - \frac{8I_{fw}}{3}. \quad (18)$$

5) Piecewise Current Expression for the First Quarter: The analysis above refers to the first quarter of the line cycle. The second quarter is a mirror of the first, and the second half-cycle is its negative counterpart. The complete piecewise expression for i_{La} in the first quarter is as follows:

$$i_{La} = \begin{cases} 0, & 0 \leq \omega t < \theta \\ \frac{V_p \sin(\omega t)}{4L_m f_s} - \frac{4I_{fw}}{3}, & \theta \leq \omega t < \frac{\pi}{3} - \theta \\ \frac{V_p \sin(\pi/3 - \theta)}{4L_m f_s} - \frac{4I_{fw}}{3}, & \frac{\pi}{3} - \theta \leq \omega t < \frac{\pi}{3} + \theta \\ \frac{V_p \sin(\omega t)}{4L_m f_s} - \frac{8I_{fw}}{3}, & \frac{\pi}{3} + \theta < \omega t < \frac{\pi}{2}. \end{cases} \quad (19)$$

Using the relationship between I_{fw} and θ from (16) we can rewrite the equation in a more concise form as given by (20) and shown in Fig. 4(a)

$$i_{La} = \begin{cases} 0, & 0 \leq \omega t < \theta \\ \frac{V_p(\sin(\omega t) - \sin(\theta))}{4L_m f_s}, & \theta \leq \omega t < \frac{\pi}{3} - \theta \\ \frac{V_p(\sin(\pi/3 - \theta) - \sin(\theta))}{4L_m f_s}, & \frac{\pi}{3} - \theta \leq \omega t < \frac{\pi}{3} + \theta \\ \frac{V_p(\sin(\omega t) - 2\sin(\theta))}{4L_m f_s}, & \frac{\pi}{3} + \theta < \omega t < \frac{\pi}{2}. \end{cases} \quad (20)$$

E. Energy Performance Analysis

This section analyzes the energy-related aspects of the IVR-PFM converter, including rms input current, harmonic distortion, ripple behavior, and input power estimation. A final comparative study consolidates these aspects by evaluating structural features against conventional Vienna Rectifier implementations.

TABLE III
STRUCTURAL AND PERFORMANCE COMPARISON: IVR-PFM AND VIENNA RECTIFIERS

Topology	# Switches/ Diodes	# Magnetics/ Windings	Switch I_{pk} (A) ^a	Switch I_{rms} (A) ^a	Switch I_{avg} (A) ^a	Switches/ Diodes Cond. Loss (W)	Switching Loss (W)	Magnetics/ Windings Loss (W)	Control Complexity	Sensors
IVR-PFM	12/ 12	6/ 9	$I_p + I_{fw}$ = 11.89 A	$(I_p + I_{fw})\sqrt{\frac{1}{8} - \frac{8V_p}{9\pi v_o}}$ = 2.94 A	$(I_p + I_{fw})\left(\frac{1}{2\pi} - \frac{V_p}{4v_o}\right)$ = 1.22 A	40 W	Negligible (soft switching)	33 W ^b	Low (only output PI)	1 voltage, half-line cycle detection
Vienna Rectifier	6/ 6	3/ 3	I_p = 11.12 A	$I_p\sqrt{\frac{1}{4} - \frac{4V_p}{3\pi v_o}}$ = 4.37 A	$I_p\left(\frac{1}{\pi} - \frac{V_p}{2v_o}\right)$ = 2.29 A	36 W	4 W ^e	7 W ^c	High	3 current, 5 voltage
Dual-phase interleaved Vienna Rectifier	12/ 12	6/ 6	$\frac{I_p}{2}$ = 5.56 A	$I_p\sqrt{\frac{1}{16} - \frac{V_p}{3\pi v_o}}$ = 2.19 A	$I_p\left(\frac{1}{2\pi} - \frac{V_p}{4v_o}\right)$ = 1.15 A	31 W	8 W ^e	10 W ^d	High	6 current, 5 voltage

Note: ^a Forward conduction only. Nominal condition: $P_o = 3$ kW, $v_o = 800$ V, $V_p = 179.6$ V, $f_s = 64$ kHz. Switches: 6R041P6; Diodes: D15S120. Magnetic Components: ^b 3x L=500 μ H (2x CS400060) + 3x Lm=52 μ H (Ferrote E30/15/14), ^c 3x 2 mH (2x CS400125), ^d 6x 1 mH (2x CS400125), ^e Switching losses negligible if SiC or GaN are used. For the present comparison, SiC diodes were adopted, and reverse recovery losses were not included.

1) **RMS Input Current:** The rms value of the input current is obtained analytically from the modeled waveform, resulting in the following closed-form expression:

$$I_{Lrms} = \frac{V_p T_s}{4L_m} \sqrt{\frac{\pi - 6(1 + \sqrt{3}\theta) \sin(2\theta) + 4(3\sqrt{3} + \pi - 3\theta) \sin^2(\theta)}{2\pi}}. \quad (21)$$

2) **THD:** The THD of the input current can be estimated as a function of θ . The rms value of the fundamental component I_{L1} is obtained from the Fourier series of i_{La} , which contains only sine terms due to odd symmetry. From this, the THD is calculated as the ratio between the total rms current and its fundamental

$$I_{L1} = \frac{V_p T_s}{4L_m} \cdot \frac{\pi - 6\theta}{\pi\sqrt{2}} \quad (22)$$

$$\begin{aligned} \text{THD}(\theta) &= \sqrt{\left(\frac{I_{Lrms}}{I_{L1}}\right)^2 - 1} \\ &= \sqrt{\frac{\pi(\pi - 6(1 + \sqrt{3}\theta) \sin(2\theta) + 4(3\sqrt{3} + \pi - 3\theta) \sin^2(\theta))}{(\pi - 6\theta)^2}} - 1. \end{aligned} \quad (23)$$

Equation (23) captures the main distortion behavior, which is nearly linear for typical values of θ . A simple approximation $\text{THD} \approx 0.6\theta$ fits well, as shown in Fig. 4(b).

While the analytical domain allows $0 < \theta < \pi/6$, practical implementations generally yield low θ values due to commercial component characteristics. However, as defined in (16), θ is also influenced by design parameters, and improper choices may result in impractically high values—leading to increased distortion and higher I_{fw} , with associated conduction losses.

3) **Input Current Ripple:** Fig. 5 shows the parametrized input current ripple profile for different modulation index values ($M = v_o/V_p$) ranging from 4 to 10. Unlike conventional analysis, where the input current ripple is defined by a single equation, this converter's ripple is determined by the highest voltage drop across the input inductor during each stage, as shown in Table II. Each stage involves three possible equations based on the phase angle and voltage gain, leading to nine unique ripple equations overall, which are reflected in Fig. 5.

4) **Input Power Estimation:** The average input power P_i is calculated from the analytical current waveform and

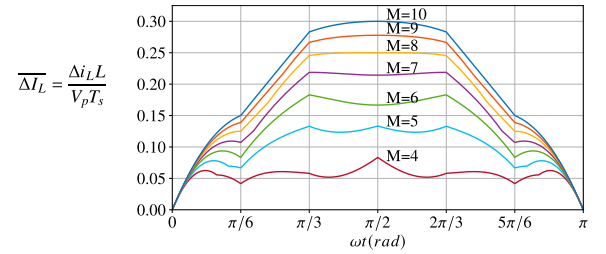


Fig. 5. Input current ripple parametrized.

voltage-follower operation, resulting in

$$P_i = \frac{3V_p^2}{8L_m f_s} \left(1 - \frac{6\theta}{\pi}\right) = \frac{3V_p^2}{8L_m f_s} - \frac{12V_p I_{fw}}{\pi}. \quad (24)$$

Here, θ is considered small, allowing $\sin(\theta) \approx \theta$ for simplification.

5) **Comparative Structural and Performance Analysis:** To better contextualize the performance and tradeoffs of the IVR-PFM, Table III presents a comparative overview of its structural and functional characteristics against two Vienna Rectifier implementations: the conventional version, as shown in Fig. 1(a), and a dual-phase interleaved variant, composed of two Vienna Rectifiers operating in parallel with interleaved control. The table includes estimates of switch current stress, conduction and magnetic losses, as well as control complexity and sensing requirements.

The IVR-PFM exhibits higher current levels in the switches, which result from circulating currents inherent to its operation. These currents increase both conduction losses in semiconductors and core losses in the magnetics. All values were obtained under the same nominal conditions and using the same component models listed in the table footnote.

While this leads to higher total losses compared to conventional Vienna Rectifiers, the IVR-PFM achieves soft switching across the full load range, making switching losses negligible and potentially reducing EMI—though not measured in this study. Hard-switched topologies such as the Vienna Rectifier, on the other hand, can leverage wide bandgap devices to minimize switching losses, albeit at the cost of greater control and sensing complexity. The IVR-PFM, by contrast, operates with a simpler voltage-only control scheme and does not require current sensing.

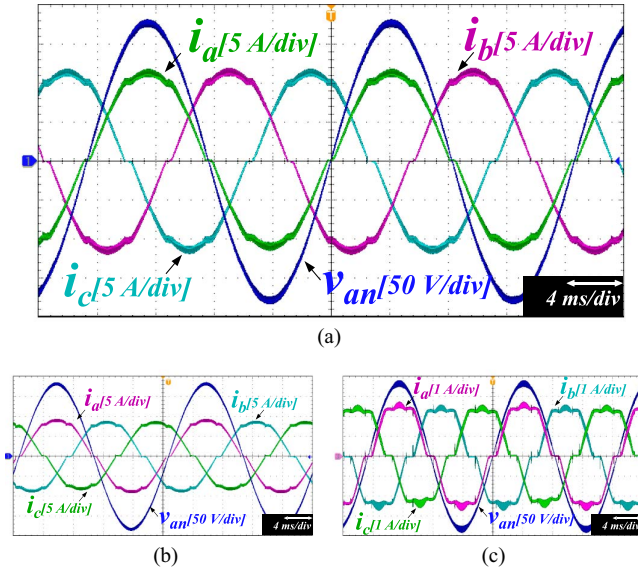


Fig. 8. Input phase voltage and three-phase currents waveforms. Input current follows the input voltage, showing resistive behavior. Load: (a) nominal; (b) 80%; and (c) 25%.

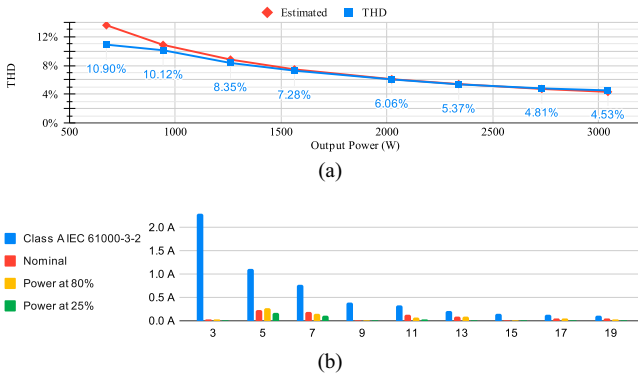


Fig. 9. (a) THD across different load conditions and analytical estimation (23). (b) Harmonic comparison with IEC 61000-3-2 Class A.

input current waveform quality, with THD consistently decreasing as power increases. A more detailed harmonic analysis is presented in Fig. 9(b), which shows the input current harmonic spectrum for nominal load, 80%, and 25% of nominal load. Although the visible flattening near the zero-crossing points at lower power levels is evident, the measured harmonic content for all operating conditions remains within the limits specified by IEC 61000-3-2 Class A for equipment above 1 kW. This supports compliance at reduced loads. Using the analytical relation (23) ($\text{THD} \approx 0.6 \theta$) and the θ –power dependence in (16), a conservative extrapolation places the 25% THD reference near $P_i \approx 0.04 P_{\text{rated}}$ for this design (region not exercised); all measured points ($\leq 25\%$ load) remained below 11%.

The free-wheeling current ($I_{fw} = 0.734$ A) and zero-crossing deviation ($\theta = 0.074$ rad), estimated under nominal operating conditions, were calculated using selected intrinsic capacitance values of $C_s = 400$ pF for the switches and $C_d = 300$ pF for the diodes. While datasheets provide

useful starting points, their values are typically influenced by reverse voltage, temperature, and other application-specific conditions. Additional contributions from PCB layout parasitics and construction details also affect the actual behavior. Nevertheless, the precise knowledge of these parameters is not required for the converter to operate properly or for its control implementation. From a design perspective, however, their approximate estimation is recommended to better align the converter's nominal behavior and efficiency with the intended operating range.

Beyond the values considered in the model, some additional nonideal effects are known to influence the behavior of real implementations and are worth highlighting. Within typical tolerances, small constructive variations in L_m do not unbalance operation or significantly alter the expected behavior.

At the device level, the intrinsic capacitances of the switches and diodes are not ideal linear elements. These capacitances originate from the devices' internal physical structure and exhibit different behaviors during charging and discharging cycles. Their nonlinear and asymmetric nature alters the timing and shape of voltage transitions, especially during resonant switching events, and can shift the effective I_{fw} compared to its idealized prediction.

Furthermore, the printed circuit board introduces stray inductances and parasitic capacitances that are difficult to characterize precisely. These layout-related effects influence the amplitude and duration of resonant transitions and affect the formation of the free-wheeling current.

Another relevant point concerns the behavior of the body diodes in the power switches. For example, in CoolMOS devices, the body diode is not as fast as ultra-fast or SiC types. When it transitions from reverse bias to conduction, the lightly doped drift region initially presents high resistance. As current begins to flow, charge modulation gradually reduces this resistance, but during this brief interval, the forward voltage rises above the typical steady-state value. This behavior, described in [25, pp. 74], leads to transient increases in conduction losses—particularly relevant at higher switching frequencies, where such short conduction intervals occur more frequently.

Fig. 10 shows measured waveforms during a switching cycle at $f_s = 90$ kHz, highlighting the main commutation stages of the converter. Fig. 10(a) presents the input current i_{La} , the voltage v_{o1} across capacitor C_{o1} , the voltage v_{ab} between the autotransformer windings, and the interlaced current measured between windings of the autotransformer. This measured current is interpreted as the theoretical magnetizing current i_{Lma} , which, according to the analytical model, is equal to twice the current i_{4Lma} flowing through the equivalent inductance $4L_{ma}$. The waveform reflects the expected dynamics: the current follows the polarity changes of v_{ab} and reaches peak values consistent with the sum of i_{La} and two times the free-wheeling current I_{fw} during commutation intervals.

Fig. 10(b) complements this view by showing the current and voltage of switch S_{1a} and diode D_{1a} . The current waveforms share the same dynamic profile imposed by the magnetizing inductance: increasing and stabilizing in the switch, and decreasing in the diode. The commutation events are partially visible

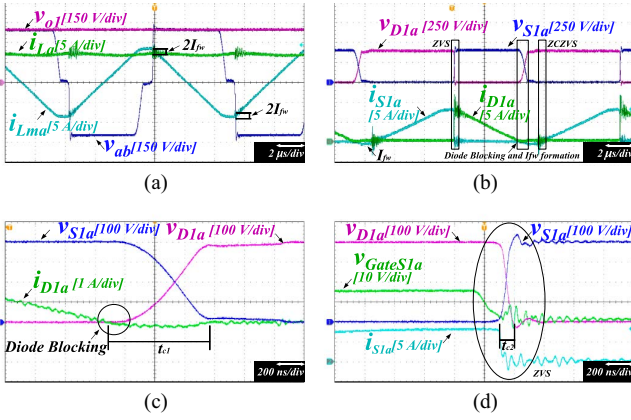


Fig. 10. Experimental waveforms during a switching cycle, $f_s = 90$ kHz. (a) Converter operation: i_{La} , voltage across windings v_{ab} , C_{o1} output voltage v_{o1} , and interleaved winding current approximating the theoretical magnetizing current i_{Lma} , which accounts for twice the free-wheeling current I_{fw} . (b) Commutation of S_{1a} and D_{1a} showing ZVS and ZVZCS conditions. (c) Diode blocking: i_{D1a} reaches zero, v_{D1a} rises; t_{c1} marks onset of I_{fw} . (d) Fast turn-off of S_{1a} due to $i_{La} + I_{fw}$, enabling ZVS. t_{c2} annotated.

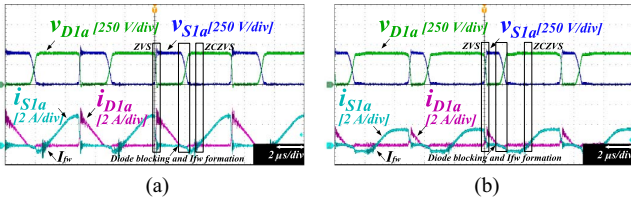


Fig. 11. Commutation waveforms of S_{1a} and D_{1a} at $f_s = 200$ kHz, under low power conversion. (a) Near peak input current. (b) Shortly before input current reaches zero. All soft-switching transitions are maintained: diode blocking with zero current, ZVS during diode conduction, and ZVZCS at switch turn-on.

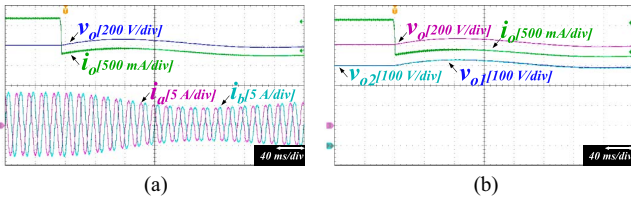


Fig. 12. Step response of the IVR-PFM converter under load changes. (a) Output voltage and current response, with two input currents shown, indicating stable operation with minimal overshoot and rapid settling. (b) Output voltage across the two output capacitors, demonstrating balanced voltage without requiring a dedicated voltage-balancing control loop.

in this time scale, with the diode turning OFF at zero current, the switch turning OFF under ZVS, and ZVZCS conditions present during the turn-ON.

To observe the same behavior under reduced current levels, Fig. 11 shows waveforms captured at $f_s = 200$ kHz. The two sets of measurements, shown in Fig. 11(a) and (b), correspond to instants near the peak and near the lower portion of the input current cycle. In both cases, the same commutation patterns are observed: diode blocking occurs with zero current, and the

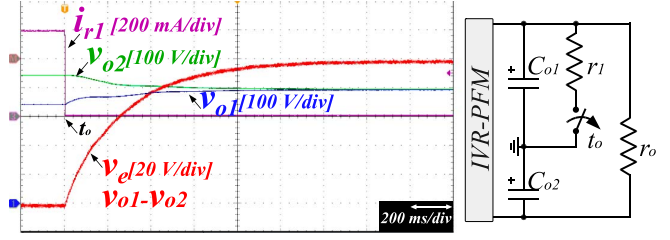


Fig. 13. Voltage-balance of the output capacitors, the IVR-PFM does not need additional output voltage balance control loop.

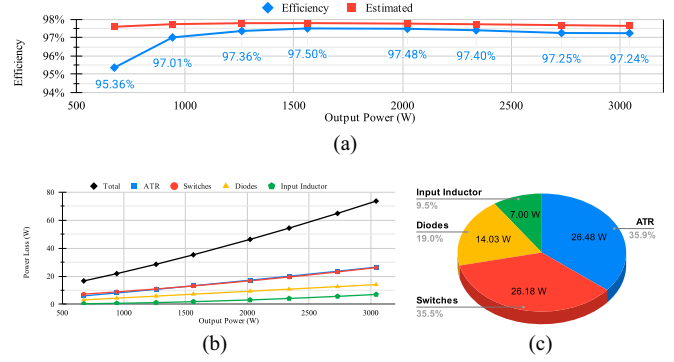


Fig. 14. (a) Efficiency across different load conditions. And loss breakdown: (b) as a function of the output power and (c) at nominal condition.

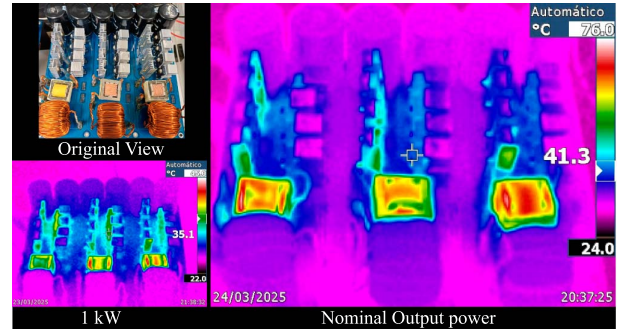


Fig. 15. IVR-PFM thermal pictures at nominal power (3 kW) and partial power (1 kW).

transitions remain soft, with ZVZCS and ZVS conditions identifiable. These results demonstrate that soft-switching operation is maintained across a wide current range.

Fig. 10(c) and (d) focusses on the detailed behavior of two critical transitions. In Fig. 10(c), the blocking of D_{1a} occurs as its current reaches zero, followed by a slow voltage rise driven by the resonance between the intrinsic capacitances of the diode and switch, and the equivalent inductance $4L_m$. This interval is marked as t_{c1} , and it coincides with the formation of the free-wheeling current I_{fw} .

Fig. 10(d) shows the transition when S_{1a} turns OFF and D_{1a} begins to conduct. The gate signal, current through the switch, and reverse voltages of S_{1a} and D_{1a} are presented. The sum of i_{La} and I_{fw} contributes to a fast and smooth voltage transition, enabling ZVS. The commutation time t_{c2} is annotated and,

TABLE V
COMPARISON: THREE-PHASE RECTIFIERS

Reference	Sensors	Magnetic components	Semiconductor components	Sensorless	Soft commutations; interleaved	Control and processor requirements	Efficiency	THD	V_L , V_o , P_o	Comments
[3]	3 current 1 voltage	3 cores 3 winding	6 diodes 6 switches	Input voltage	~	Inner and outer PI control loop with PWM	98.56%	3.6%	110 V, 200 V, ~	Proposes a sensorless control technique for a front-end rectifier in telecom systems, regulating output voltage without input voltage sensors. It uses PI controllers for both the outer voltage loop and the inner current loop to achieve power factor correction and maintain low THD
[7]	4 voltage	3 cores 3 windings	18 diodes 3 switches	Input current	~	$\alpha\beta$ transformation, cosine calculation and outer PI control loop with PWM	~	~	60 V, 100 V, ~	Proposes a current sensorless control strategy for unity power factor operation. While efficiency and THD evaluations are not provided, the unity power factor is evident in the experimental results. The control strategy is simple, using only an outer-loop PI, but requires cosine calculation and $\alpha\beta$ transformation, potentially increasing processing demands.
[8]	3 current 1 voltage	3 cores 3 winding	6 diodes 6 switches	Input voltage	~	SOGI, FLL, and outer PI control loop with SVPWM	~	$\approx 2\%$ unbalanced grid conditions	208 V, 300 V, ~	Proposes a sensorless control strategy that eliminates the need for grid voltage sensors by using harmonic estimation for control, with the Vienna rectifier driven by SVPWM. It effectively reduces input current harmonics under unbalanced and distorted grid conditions. However, the voltage estimation and control process are complex and may require significant computational power.
[12]	3 current 5 voltage	3 cores 3 windings	6 diodes 6 switches	~	~	$\alpha\beta$, dq0, inner predictive control loop and outer PI control loop	$\approx 98.3\%$	2.29%	220 V, 340 V, 4 kW	Proposes a hybrid control scheme combining PI as the outer control with finite set-model predictive control (FS-MPC). The method requires $\alpha\beta$ and dq transformations. The article demonstrates good output voltage balance and regulation of input reactive power.
[15]	3 current 1 voltage	6 cores 9 windings	12 diodes 12 switches	~	Interleaved	Discontinuous Space-Vector Modulation	99.28%	9.5%	230 V, 650 V, 3 kW	Focuses on designing an interleaved Vienna rectifier optimized for maximum efficiency. Both use ATR (interphase inductors) for current sharing, but ours also leverages the circulating current for resistive behavior.
IVR-PFM	1 voltage half-line cycle detection	6 cores 9 winding	12 diodes 12 switches	Input current and voltage	ZVS, ZVZCS, no reverse recovery; Interleaved	Outer PI control loop and PFM	97.50%	4.53%	220 V, 800 V, 3 kW	Proposes a novel topology integrating the SC-PFM and Vienna Rectifier for three-phase applications. It achieves high efficiency and unitary power factor without current sensors, leveraging soft commutations and resistive behavior for simplified control. The method reduces system complexity by eliminating dq0, $\alpha\beta$, and cosine processing while maintaining continuous input current and high static gain.

being faster than t_{c1} , it may be neglected in the input current equation for the sake of simplicity.

Fig. 12 illustrates the converter's step response under a load change. The output voltage and current respond quickly, with minimal overshoot and short settling time, as the converter adapts through adjustments in the switching frequency. During this transient, the switching period T_s exhibited a maximum slew rate of $34 \mu\text{s/s}$ and a peak deviation of 16% relative to the final steady-state value.

To experimentally verify the voltage balancing behavior, an unbalanced condition was created by connecting an additional resistor of approximately 870Ω in parallel with capacitor C_{o1} , reducing its voltage relative to C_{o2} . The total output resistance across both capacitors (v_o) was approximately 316Ω . Even under this unbalanced loading, the converter maintained stable operation without significant disturbance in the overall output voltage regulation. After reaching steady state, the additional resistor was disconnected via a switch. The resulting voltage difference decayed exponentially, as shown in Fig. 13, confirming the natural balancing effect of the topology without requiring a specific control loop for voltage balancing.

It is worth noting that, in the experimental setup, the implementation followed the control structure shown in Fig. 6, but with some practical adaptations. Voltage transducers (LEM LV 25-P) were used for output sensing, and due to their 500 V limit, two sensors were applied—one for each output capacitor. For half-line cycle detection, three voltage sensors were used to measure v_{an} , v_{bn} , and v_{cn} , with digital processing handling the detection in the DSP. These choices reflect implementation constraints and do not alter the control structure, which remains as illustrated in Fig. 6 and can be adjusted according to design preferences.

Fig. 14(a) shows the converter's efficiency across different load conditions. The efficiency curve demonstrates that the converter maintains high efficiency over a wide range of output power, as well as the loss breakdown given in Fig. 14(b) and (c).

This high efficiency is attributed to the few components in the current path due to the Vienna Rectifier layout and the soft commutations provided by the SC-PFM, both of which help reduce losses during operation.

Fig. 15 presents the thermal behavior of the converter at two operating points: nominal power (3 kW) and partial load (1 kW). The power dissipation is mainly concentrated in the autotransformer, switches, and diodes. Simple heat sinks were used for thermal management and proved adequate during the tests. For commercial designs, however, further thermal optimization is advisable. In particular, the autotransformers exhibit the highest thermal concentration and could benefit from improved core selection, winding design, or dedicated cooling strategies.

Table V compares the proposed converter with other PFC converters from the literature, highlighting its advantages.

A simulation file for the IVR-PFM topology has been made available for replication and analysis. This dataset includes a closed-loop configuration and options for open-loop testing [26].

V. CONCLUSION

The IVR-PFM achieves high efficiency and power quality in PFC applications. Integrating SC-PFM enables continuous input current without current sensing, simplifying control and reducing complexity.

Experimental waveforms confirm its resistive behavior, with input currents closely aligned to input voltage. Despite nonidealities such as free-wheeling current and zero-crossing disturbances, the converter achieves low THD and near-unity power factor, demonstrating excellent power quality.

Efficiency measurements further highlight the benefits of the Vienna Rectifier's layout and SC-PFM's soft-switching, both contributing to reduced losses. The control strategy effectively manages switching, ensuring stable performance under dynamic load conditions.

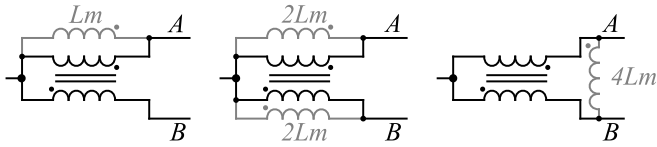


Fig. 16. Illustration of the equivalence between conventional and alternative representations of magnetizing inductance. (a) Conventional representation. (b) Intermediate representation where the inductance is doubled ($2L_m$) and present in both windings simultaneously. (c) Alternative representation, where inductor $4L_m$ is connected at points a and B.

APPENDIX

MAGNETIZING INDUCTANCE REPRESENTATIONS

To understand the nonidealities in the operation of the converter, it's beneficial to represent the magnetizing inductance of the autotransformer in an alternative manner [27]. Typically, the literature represents the magnetizing inductance as a single inductor in parallel with one winding of the transformer. However, for didactic purposes and to better explain the nonidealities, we use a different representation.

In this article, we represent the magnetizing inductance with an inductor in parallel with both connections of the autotransformer, labeled as points A and B. This inductor has an inductance value that is four times the conventional magnetizing inductance. Both representations exhibit similar behavior, provided that the windings have a 1:1 ratio. Fig. 16 presents a step-by-step illustration of how the conventional representation with one magnetizing inductor L_m in parallel with one winding can be represented with another inductor with $4L_m$ inductance, connected at points A and B.

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